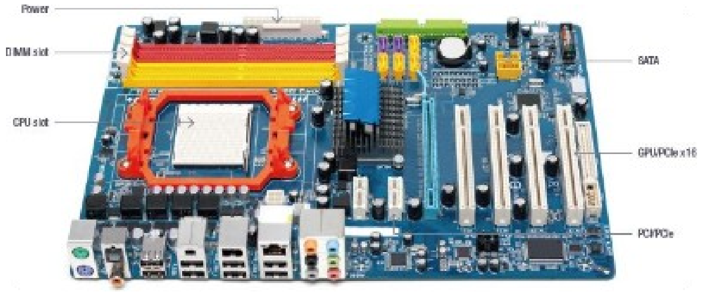
UNIT 3: MEMORY ORGANIZATION

COMPUTER ORGANIZATION



Presented By: Bibrata Sarkar
Newton School of AI & Computing

Complete Unit Roadmap

Memory Org → Hierarchy → Registers → Main Memory → RAM → ROM → Cache Memory → Locality → Mapping → Secondary Storage → Memory Mapped I/O → Isolated I/O

> Provides a structured path from fastest internal CPU memory down to permanent storage solutions.

Industry Applications

Artificial intelligence: Requires high bandwidth memory (HBM) to process neural networks rapidly.

Cloud computing: Relies on vast server RAM and SSDs for virtualization and fast data retrieval.

Autonomous Vehicles: Need real-time cache processing for instant sensor decision-making.

Gaming systems: Utilize fast GDDR memory to render high-resolution textures without lag.

Memory constraints often dictate software capabilities in the real world.

Course Objectives

Course Outcomes

Program Outcomes

CO–PO Mapping

Program Specific Outcomes

CO–PSO Mapping

Program Educational Objectives

Result Analysis Dashboard

Question Paper Blueprint

Prerequisite Knowledge

Introduction to Memory Organization

- **Why computers need memory:** A CPU executes instructions rapidly but has no large internal space to keep software code.
- **Instruction Storage:** Holds the binary code that tells the CPU what to do.
- **Execution Dependency:** Processors literally cannot function without memory to fetch from.

Storage (HDD) → Memory (RAM) → CPU Processing → Output

Learning Resources

Standard textbooks:

1. Computer Organization - Carl Hamacher
2. Computer Architecture - William Stallings

Online Video Channels:

- NPTEL Computer Architecture
- Gate Smashers
- Neso Academy

Web resources:

- GeeksforGeeks
- Official Intel Documentation

Learning Objectives

After completing this unit, students will be able to:

- Explain Memory Organization and Memory Hierarchy.
- Differentiate Registers, Cache, RAM, and Storage.
- Explain the workings of RAM, ROM, SRAM, and DRAM.
- Describe Locality of Reference and Cache Mapping Techniques.
- Explain HDD, SSD, and specialized I/O mapping methods.

Unit Contents Roadmap

1. Memory Intro
2. Hierarchy
3. Memory Characteristics
4. Primary Memory
5. RAM (SRAM/DRAM)
6. ROM Types
7. Cache Memory
8. Cache Mapping
9. Secondary Storage
10. I/O Methods

Topic: Intro to Memory Organization

TOPIC OBJECTIVE:
To understand why memory is required in a computer system and how different memory components work together to improve system performance.

TOPIC OUTCOME:
Students should be able to explain the role of memory, differentiate between storage and memory, and identify different memory categories.

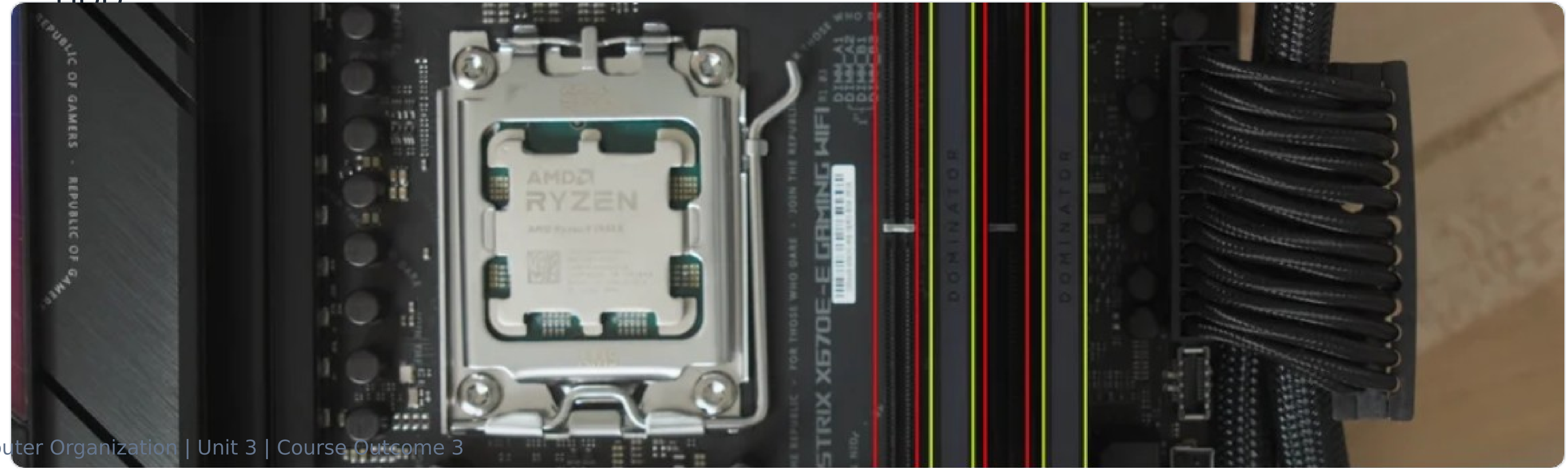
What is Memory?

"Can a computer work without memory?"

No. Memory is the working space and storage area of a computer.

Definition: Memory stores programs, instructions, data, intermediate results, and the Operating System.

- **Brain Analogy:** Short-term memory = Registers/Cache. Working memory = RAM. Long-term memory = HDD



Why Do Computers Need Memory?

- **No memory = No execution:** The CPU requires a workspace to perform calculations.
- **Application Execution:** E.g., Opening MS Word copies the software from the hard drive into fast memory so the CPU can run it.

Power ON → BIOS → RAM Init → OS Loads → App Runs

Analogy: Memory is a student's notebook. Without it, solving complex equations purely in mind is impossible.

Functions of Memory

Instruction Storage: Holds active program code.

Data Storage: Holds user files and variables.

Intermediate Storage: Keeps calculation results temporarily.

Buffering/ Caching: Smooths out speed differences between devices.

Exam Point: Understand the strict difference between active Memory (volatile) and Storage (permanent).

Classification of Computer Memory

Computer Memory

↓

Primary Memory:
Registers, Cache, RAM, ROM
(Volatile, Fast)

Secondary Memory:
Hard Disk, SSD, Optical
(Non-Volatile, Slow, Large)

Evolution of Computer Memory

Vacuum Tubes → **Magnetic Drum** → **Magnetic Core** → **Semiconductors (SRAM/DRAM)** → **NAND Flash (SSD)**

Generational Improvements:
Every generation aimed to increase Speed and Capacity while decreasing Physical Size, Power Consumption, and Cost.

Daily Quiz

Test your understanding:

- 1. What is Memory Organization?
- 2. Why is memory important for the CPU?
- 3. Differentiate between memory and storage.
- 4. Which memory stores currently running programs?
- 5. Is RAM permanent?
- 6. Name two examples of secondary storage.
- 7. Can a CPU execute programs purely from a hard drive?

Weekly Assignment

Submit written answers for the following:

- Q1. Explain the fundamental concept of Memory Organization.
- Q2. Draw the memory classification tree diagram.
- Q3. Explain the distinct functions of computer memory.
- Q4. Explain the role of memory during the computer booting sequence.
- Q5. Why is RAM referred to as temporary memory? Justify.

Topic Learning Resources

Reading
:
William Stallings - Chapter 4 (Memory)

Carl Hamacher - Chapter 8 (Memory System)

Video Links:https://www.youtube.com/watch?v=LE_lyLCqy8I&list=PL3R9-um41JszyaKeoc9qP8Bn45XzqJycj

Topic: Memory Hierarchy

TOPIC OBJECTIVE:
To understand the arrangement of different memories according to speed, capacity, and cost.

TOPIC OUTCOME:
Students should be able to explain the Hierarchy, identify every level, justify its necessity, and compare memory tiers.

Introduction to Memory Hierarchy

The Problem:

CPUs execute billions of instructions per second. No single memory device can provide the highest speed, largest capacity, and lowest cost simultaneously.

The Solution:

Computers combine multiple memory technologies into a hierarchical structure.

- **Closer to CPU:** Faster, more expensive, lower capacity.
- **Farther from CPU:** Slower, cheaper, massive capacity.

Memory Hierarchy Diagram

Registers

Cache (L1, L2, L3)

Main Memory (RAM)

Solid State Drive (SSD)

Hard Disk Drive (HDD)

↑ Speed Increases

↑ Cost/Bit Increases

↓ Capacity Increases

Detailed Explanation of Hierarchy

Memory Type	Capacity	Speed	Cost
Registers	Bytes	Fastest	Highest
Cache	KB / MB	Very Fast	Very High
RAM	GB	Fast	Moderate
SSD/HDD	TB	Slow	Lowest

Classroom Discussion: Why don't manufacturers simply build computers using only cache memory?

Topic: Characteristics of Memory

TOPIC OBJECTIVE:
To understand the major characteristics used to evaluate and compare different memory technologies.

TOPIC OUTCOME:
Students will be able to explain memory speed, capacity, cost, access time, and select appropriate memory for applications.

Characteristics of Memory Systems

Every memory technology is evaluated using measurable properties to determine its performance and efficiency.

Speed	**Capacity**	**Cost per Bit**
Volatility	**Access Method**	**Power Draw**

A memory designer must mathematically balance Speed, Capacity, and Cost for optimal system architecture.

Memory Speed

Definition The time required to read or write data from memory.

- **Access Time:** Time taken between request and delivery of data.
- **Cycle Time:** Minimum time between two independent memory accesses.
- **Transfer Rate:** Rate at which data is transferred (e.g., GB/s).

Exam Point: Understand why Registers (inside the processor chip) are inherently faster than RAM (separated by a motherboard bus).

Memory Capacity

Capacity represents the total amount of digital information the memory can store.

Bit → Byte (8) → KB → MB → GB → TB

- **RAM Capacity:** Typically 8 GB to 64 GB in standard PCs.
- **HDD Capacity:** Typically 1 TB to 16 TB.

Industrial Insight: Modern AI training servers utilize hundreds of gigabytes of specialized memory to load entire datasets.

Memory Cost

Cost is evaluated as **Cost per Bit**.

- **SRAM (Cache):** Very expensive due to complex transistor logic.
- **DRAM (RAM):** Cheaper due to simple capacitor-based design.
- **HDD:** Lowest cost per bit due to magnetic platter density.

> **Higher Speed = Higher Cost**
> **Higher Capacity = Lower Cost per Bit**

Comparison of Characteristics

Memory	Speed	Capacity	Cost	Volatile
Registers	Ultra Fast	Tiny	Highest	Yes
Cache	Very Fast	Small	Very High	Yes
RAM	Fast	Medium	Moderate	Yes
SSD	Moderate	Large	Low	No
HDD	Slow	Huge	Lowest	No

Real-Life Analogy

Student Studying Analogy:

- **Registers:** The student's brain (Immediate processing).
- **Cache:** Books open on the study desk (Fastest access).
- **RAM:** Books inside the school bag (Requires a brief moment to fetch).
- **SSD/HDD:** Books in a library across town (Takes a long time, but holds massive amounts of data).

Daily Quiz

Test your understanding:

- 1. Define Access Time.
- 2. Which memory characteristic determines total storage size?
- 3. Which level of memory has the highest cost per bit?
- 4. Which memory has the largest capacity in a standard PC?
- 5. Why is speed a critical evaluation metric?
- 6. Can ultra-low-cost memory also be ultra-fast? Why?

Weekly Assignment & Resources

Assignment:

- Explain all memory characteristics.
- Prepare a comparison table.
- Draw a graph of Cost vs. Speed.
- Explain the necessity of mixed technologies.

Resources:

https://www.youtube.com/watch?v=LE_lyLCqy8I&list=PL3R9-um41JszyaKeoc9qP8Bn45XzqJycj

Topic: Primary Memory

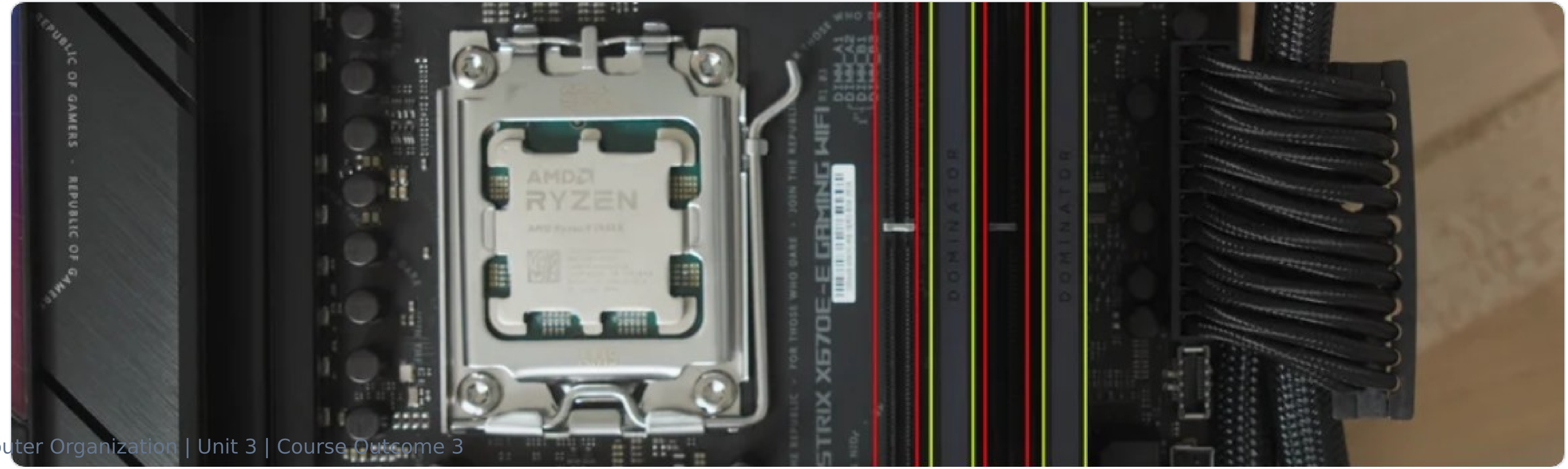
TOPIC OBJECTIVE:
To understand Primary Memory and its role in storing active programs and data.

TOPIC OUTCOME:
Students will be able to explain Main Memory, describe RAM, differentiate RAM and ROM, and identify its architectural placement.

Introduction to Primary Memory

Definition: Directly accessible by the CPU, it stores currently executing programs and active data.

- Also known as Main Memory.
- Acts as the working area of the computer.
- CPUs cannot directly execute code stored on a Hard Drive; it must be moved here first.



Features of Primary Memory

Direct CPU Access: Communicates via address and data buses.

Fast Speed: Electronic, no moving parts.

Volatile (Mostly): RAM loses data on power off.

Supports Multitasking: Holds multiple active apps simultaneously.

Example: Opening PowerPoint allocates a chunk of Primary Memory to hold the application's code and your slides.

Working of Primary Memory

1. Power ON

2. Operating System loads from SSD into RAM

3. User launches an application; it loads into RAM

4. CPU fetches instructions sequentially from RAM

5. App closes; RAM space is freed up for other tasks

Primary Memory Architecture

CPU

↕ **(Buses)**

MAIN MEMORY (RAM)

Address Bus: Selects the specific memory location.

Data Bus: Carries the actual binary data.

Control Bus: Signals Read or Write operations.

RAM vs ROM (Overview)

Feature	RAM (Random Access Mem)	ROM (Read Only Mem)
Volatility	Volatile (Loses data)	Non-Volatile (Keeps data)
Operation	Read and Write	Mostly Read Only
Purpose	Active programs/data	Firmware, BIOS, Boot code
Speed	Very Fast	Slower than RAM

Practical Classroom Example

Scenario: Opening Google Chrome

- **SSD:** Stores Chrome permanently when PC is off.
- **OS:** Copies required Chrome files into RAM.
- **RAM:** Holds the active tab data.
- **CPU:** Fetches instructions to render the webpage.

Discussion: Why does Chrome reopen slowly after a PC restart, but quickly if you just closed it a minute ago? (Hint: OS RAM Caching).

Topic: Random Access Memory (RAM)

TOPIC OBJECTIVE:
To understand the architecture, characteristics, working principle, and applications of RAM.

TOPIC OUTCOME:
Students will be able to define RAM, explain "Random Access", describe its working principle, and identify applications.

Introduction to RAM

RAM Volatile semiconductor memory that temporarily stores currently processed programs.

Why "Random Access"?

Any memory location can be accessed directly in the exact same amount of time, regardless of its physical position (unlike sequential tapes).

Working Principle of RAM

RAM operates as the CPU's immediate workspace.

- **Allocation:** OS allocates a block of addresses for a new task.
- **Fetching:** CPU requests an address; RAM outputs the data instantly.
- **Storing:** CPU computes a result and writes it back to a RAM address.
- **Deallocation:** When task ends, addresses are marked free.

RAM never executes code. It strictly serves as rapid storage for the CPU.

Internal Organization of RAM

Memory Cell Array: Grid of cells storing bits (0 or 1).

Address Decoder: Translates binary address bus signals into a specific row/col selection.

Sense Amplifier: Reads weak electrical signals from cells and boosts them.

Read/Write Circuit: Manages the directional flow of data to the Data Bus.

Read & Write Operations

READ OPERATION

1. CPU sends address via bus.
2. Control bus signals 'Read'.
3. Decoder selects cell.
4. Data sent to CPU via data bus.

WRITE OPERATION

1. CPU sends address via bus.
2. CPU places new data on data bus.
3. Control bus signals 'Write'.
4. Selected cell updates its state.

Static RAM (SRAM)

Static RAM uses complex flip-flop circuits (usually 6 transistors per cell) to store each bit.

- Retains data as long as power is on.
- **No refresh required.**
- Extremely fast.
- Low density (physically large).
- Expensive. Used primarily for Cache.

Dynamic RAM (DRAM)

Dynamic RAM stores bits as electrical charges inside microscopic capacitors.

- Capacitors leak charge naturally.
- **Requires periodic refreshing** (thousands of times per second) to retain data.
- Slower than SRAM due to refresh cycles.
- High density (1 transistor + 1 capacitor per cell).
- Cheap to mass-produce. Used for Main Memory.

SRAM vs DRAM

Parameter	SRAM	DRAM
Storage Element	Flip-Flops (Transistors)	Capacitors
Refresh Needed?	No	Yes (Constant)
Speed	Very Fast	Slower
Density / Cost	Low Density / High Cost	High Density / Low Cost
Primary Application	CPU Cache Memory	System Main Memory

Daily Quiz & Activity

Quiz

- Expand SRAM and DRAM.
- Which memory relies on capacitors?
- Which memory requires refreshing?
- Why is SRAM used for Cache?

Activity: Map the Memory Hierarchy analogy (Notebook, Desk, Library) specifically to SRAM, DRAM, and HDD.

Topic: Read Only Memory (ROM)

TOPIC OBJECTIVE:

To understand Read Only Memory, its characteristics, and its different types.

TOPIC OUTCOME:

Students will be able to explain ROM, differentiate it from RAM, describe PROM/EPROM/EEPROM, and identify practical applications.

Introduction to ROM

ROM (Read Only Memory): Non-volatile memory that permanently stores crucial instructions for system startup (Firmware).

- Retains data without power.
- Historically, data was written during factory manufacturing.
- Used for BIOS, UEFI, and embedded microcontroller programming.

Types of ROM

ROM (Masked)

PROM (Programmable)

EPROM (Erasable)

EEPROM (Electrically Erasable)

Evolution moved from strictly factory-programmed chips to user-programmable and electrically rewritable chips.

PROM, EPROM, & EEPROM

PROM: Blank chip that can be programmed exactly ONCE using a special tool (blowing internal fuses).

EPROM: Can be erased by exposing a quartz window on the chip to Ultraviolet (UV) light. Time consuming.

EEPROM: Can be erased and rewritten electrically byte-by-byte. Forms the basis of modern Flash storage.

RAM vs ROM (Practical View)

Boot Process Example:

- **Power ON:** RAM is totally empty. CPU has no instructions.
- **ROM steps in:** CPU is hardwired to read the ROM BIOS chip first.
- **Hardware Check:** ROM executes POST (Power On Self Test).
- **Handoff:** ROM instructs the system to load the OS from the HDD into the RAM.
- **Operation:** RAM takes over as the active workspace.

Topic: Cache Memory

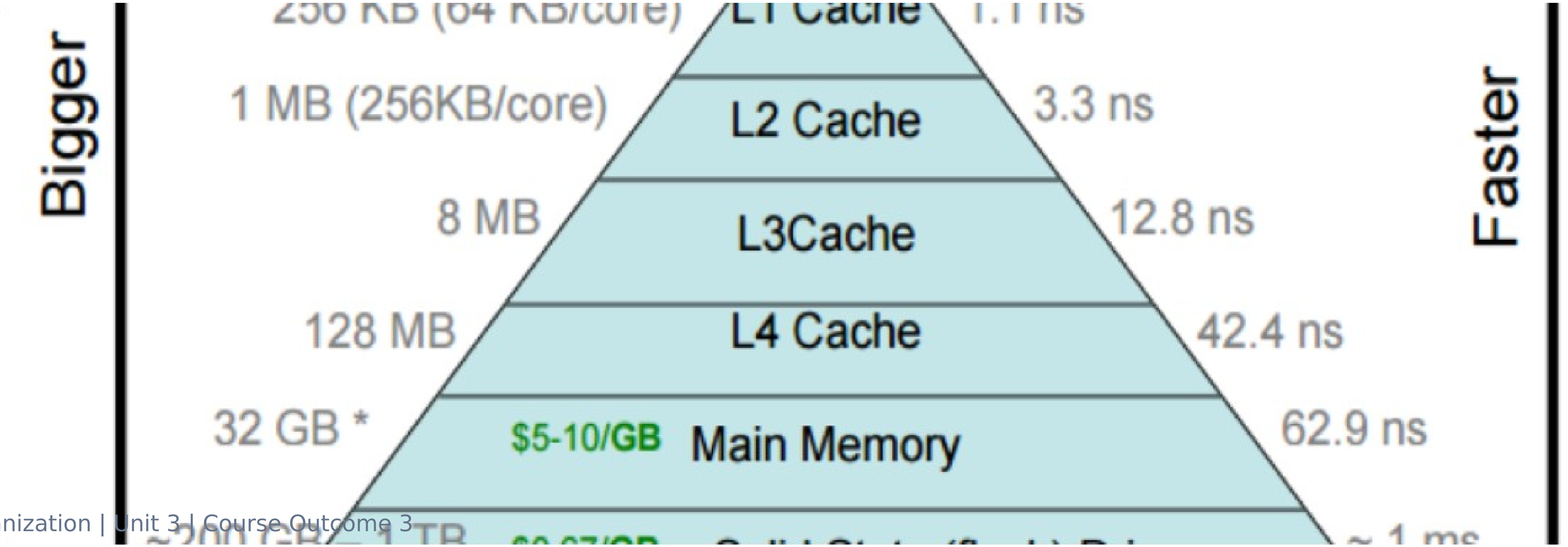
TOPIC OBJECTIVE:
To understand the purpose, architecture, and importance of Cache Memory in processor performance.

TOPIC OUTCOME:
Students will be able to define Cache, explain Cache Hit/Miss, identify Cache levels, and understand its relation to Main Memory.

Introduction to Cache Memory

Definition: A very high-speed SRAM memory located inside or adjacent to the CPU to store frequently accessed data.

The Bottleneck: CPU speeds (GHz) evolved much faster than RAM speeds (MHz). The CPU wastes time waiting for RAM. Cache bridges this speed gap.



Need for Cache Memory

Without Cache:

- CPU requests data from RAM.
- Long latency. CPU idles (Stalls).
- System performance drops.

With Cache:

- CPU finds frequent data in Cache.
- Zero latency. Execution continues immediately.
- High throughput.

Analogy: A chef (CPU) keeps spices on the counter (Cache) instead of walking to the pantry (RAM) for every pinch of salt.

Characteristics of Cache Memory

Uses **SRAM** tech.

Volatile storage.

Extremely **Fast**.

Physically **Small** capacity.

Very **Expensive** per bit.

Operates invisibly to the user.

Exam Point: Cache does not increase the total memory of a PC; it only acts as a high-speed buffer.

Cache Memory Architecture

CPU Core ↔ L1 Cache ↔ L2 Cache ↔ L3 Cache (Shared) ↔ RAM

- **L1 (Level 1):** Smallest (e.g., 64KB), inside the core, fastest.
- **L2 (Level 2):** Larger (e.g., 512KB), slightly slower.
- **L3 (Level 3):** Largest (e.g., 16MB), shared across multiple CPU cores.

Working of Cache Memory

When the CPU needs to read a memory address:

1. Hardware checks if the address is currently stored in Cache.
2. **If YES (Cache Hit):** Data is read at processor speed.
3. **If NO (Cache Miss):** Hardware halts the CPU, fetches data from RAM, copies it to Cache, and delivers it to CPU.

The goal of computer architecture is to maximize the **Hit Ratio**.

Levels of Cache Memory

Level	Location	Speed	Capacity (Typical)
L1	Inside Core	Instant	32KB - 128KB
L2	On Chip	Very Fast	256KB - 2MB
L3	On Chip (Shared)	Fast	4MB - 64MB

Modern processors use hierarchical cache so that if L1 misses, L2 is checked before taking the huge penalty of checking RAM.

Cache Hit vs Cache Miss

Cache Hit: The requested data is found in the cache.

Time taken: ~1 to 3 nanoseconds.

Cache Miss: Data is NOT found. Must access Main Memory (Miss Penalty).

Time taken: ~50 to 100 nanoseconds.

Hit Ratio = (Cache Hits) / (Total Memory Requests)

Daily Quiz

Test your understanding:

- 1. Why is Cache Memory necessary?
- 2. What semiconductor technology builds Cache?
- 3. Differentiate L1 and L3 Cache.
- 4. Define a Cache Hit.
- 5. What happens during a Cache Miss?
- 6. Why can't we make a 16GB L1 Cache?

Weekly Assignment & Resources

Assignment:

- Draw the CPU-Cache-RAM architecture.
- Explain the Memory Bottleneck.
- Define Hit Ratio and Miss Penalty.

Resources: Geek for geeks

Topic: Locality of Reference

TOPIC OBJECTIVE:
To understand the principle of Locality of Reference and its importance in cache performance.

TOPIC OUTCOME:
Students will be able to define Locality, explain Temporal and Spatial Locality, and relate it to Cache efficiency.

Introduction to Locality of Reference

Definition: The tendency of processor programs to access the same memory locations, or nearby locations, repetitively within a short time.

Programs do not access memory completely randomly. They execute in predictable patterns (like loops and arrays).

Because of Locality, Cache Memory works. By keeping recently and nearby used data in Cache, the Hit Ratio soars.

Types of Locality of Reference

1. Temporal Locality (Time)

If a memory location is referenced, it will likely be referenced again soon.

E.g., Variables in a FOR loop.

2. Spatial Locality (Space)

If a memory location is referenced, locations with nearby addresses will likely be referenced soon.

E.g., Array traversal.

Practical Examples of Locality

Code Example:

```
// Example: Matrix-Vector Multiplication (Demonstrating Locality)
#define N 1024
int matrix[N][N];
int vector[N];
int result[N];

for (int i = 0; i < N; i++) {
    for (int j = 0; j < N; j++) {
        // Temporal Locality: The value of `result[i]` is accessed
        // and updated repeatedly in a very short time span.
        //
        // Spatial Locality: `matrix[i][j]` accesses adjacent
        // elements in the row linearly, and `vector[j]` moves
        // sequentially as `j` increments.
        result[i] += matrix[i][j] * vector[j];
    }
}
```

Temporal: The variables i and sum are read/written 100 times. Cache keeps them instantly available.

Spatial: Elements array[0], array[1]... are contiguous. Cache fetches whole blocks, so array[1] is already in cache when needed!

Daily Quiz & Activity

Quiz

- Define Temporal Locality.
- Define Spatial Locality.
- How does Locality improve Hit Ratio?

Classroom Activity: Identify daily life locality. (E.g., Temporal: Using your favorite app daily. Spatial: Reading a book page-by-page).

Topic: Cache Mapping Techniques

TOPIC OBJECTIVE:
To understand how blocks of Main Memory are placed into Cache Memory.

TOPIC OUTCOME:
Students will be able to differentiate Direct, Associative, and Set-Associative mapping, understanding their pros and cons.

Introduction to Cache Mapping

Main Memory is huge (GBs). Cache is tiny (MBs). Data is transferred in "Blocks".

The Question:

When a block is copied from RAM, exactly where in the Cache should it go?

Cache Mapping is the algorithm hardware uses to determine a block's placement in Cache.

Three Types: Direct, Associative, Set-Associative.

Direct Mapping

Rule: A Main Memory block can be placed in exactly ONE specific Cache line.

Formula: Cache Line = (Main Memory Block) MOD (Total Cache Lines)

- **Advantage:** Very simple hardware, extremely fast to check if data is there.
- **Disadvantage:** *Conflict Misses.* If Block 0 and Block 8 both map to Line 0, they will constantly overwrite each other even if the rest of the cache is empty.

Direct Mapping Example

Assume 8 Cache Lines (0 to 7).

- Block 0 maps to Line 0.
- Block 1 maps to Line 1.
- ...
- Block 7 maps to Line 7.
- **Block 8 maps to Line 0.** (8 mod 8 = 0)

If a program needs Block 0 and Block 8 repeatedly, they will continuously evict each other.

Associative Mapping

Rule: A Main Memory block can be placed in ANY available Cache line.

- **Searching:** Hardware must check the tags of ALL cache lines simultaneously to find data (Parallel Search).
- **Advantage:** Maximum flexibility. Zero conflict misses. Memory blocks are only evicted when the cache is 100% full.
- **Disadvantage:** Requires extremely complex, expensive hardware (comparators for every line).

Associative Mapping Example

Analogy: An Open Parking Lot

In Direct Mapping, your car (Block) is assigned a specific parking spot (Line). If someone is in it, you can't park, even if 99 other spots are empty.

In Associative Mapping, you can park in ANY empty spot. However, the parking attendant (Hardware) has to search the entire lot simultaneously to find your car later.

Set-Associative Mapping

The Compromise: Best of both worlds. Used in modern processors.

Cache is divided into *Sets* (e.g., 2 lines per set).

- A memory block maps directly to a specific **Set**.
- Inside that Set, it can be placed in **Any Line** (associative).
- **Advantage:** Balances hardware complexity and conflict misses effectively.

Set-Associative Mapping Example

Assume an 8-line cache, configured as **2-Way Set Associative** (4 Sets total, 2 lines per set).

- Block 0 maps to **Set 0**.
- It can sit in Line A or Line B of Set 0.
- Block 4 maps to **Set 0**.
- Block 8 maps to **Set 0**.

If Block 0 and Block 4 are needed, they can BOTH reside in Set 0. Only when Block 8 is added does an eviction happen.

Comparison of Mapping Techniques

Feature	Direct	Associative	Set-Associative
Placement	Fixed Line	Anywhere	Fixed Set, Any Line
Conflict Misses	High	None	Low
Hardware	Simple/Cheap	Complex/Expensive	Moderate
Speed of Check	Fastest	Slower (parallel)	Fast
Modern Usage	Rare	Rare (TLBs)	Standard (L1/L2/L3)

Daily Quiz & Weekly Assignment

Quiz:

- Which mapping technique has the simplest hardware?
- Which mapping technique is widely used today?

Assignment: Prepare a comparison table of mapping techniques and explain the trade-off between conflict misses and hardware complexity.

Topic: Secondary Storage

TOPIC OBJECTIVE:
To understand Secondary Storage devices, characteristics, and importance.

TOPIC OUTCOME:
Students will be able to explain secondary storage, differentiate it from primary memory, and compare HDD vs SSD.

Introduction to Secondary Storage

Definition: Non-volatile memory used to permanently store programs, OS, and user data.

- Because RAM is volatile (erases on power loss), computers require permanent archives.
- **Characteristics:** Massive capacity (Terabytes), lower cost per bit, but much slower than RAM.
- **Examples:** Hard Disk Drive (HDD), Solid State Drive (SSD).

Primary vs Secondary Storage

Feature	Primary (RAM)	Secondary (HDD/SSD)
Volatility	Volatile	Non-Volatile
CPU Access	Direct via Address Bus	Indirect (via I/O controllers)
Speed	Nanoseconds (Fast)	Milliseconds (Slow)
Usage	Running Applications	Permanent File Archives

Hard Disk Drive (HDD)

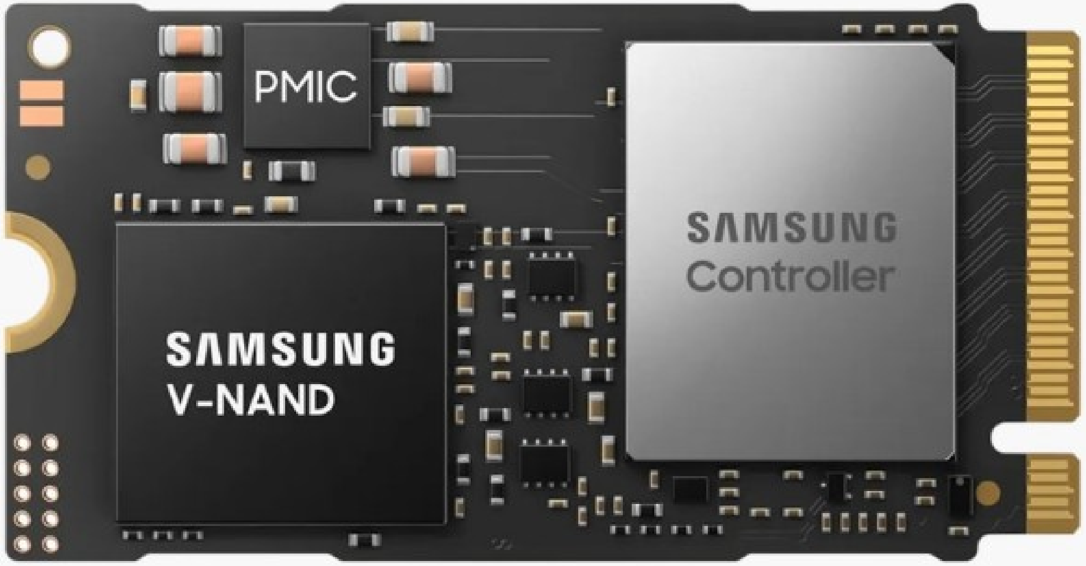
Uses spinning magnetic platters and moving mechanical read/write heads.

- **Pros:** Massive storage capacity, very low cost, reliable long-term archive.
- **Cons:** Mechanical parts cause latency, sensitive to physical shock, noisy, high power draw.

Solid State Drive (SSD)

Uses NAND Flash Memory chips to store data electronically.

- **No moving parts.**
- **Pros:** Extremely fast boot/load times, silent, shock-resistant, low power.
- **Cons:** Higher cost per GB, finite write cycles (though sufficient for consumers).



HDD vs SSD Comparison

Parameter	Hard Disk Drive (HDD)	Solid State Drive (SSD)
Technology	Magnetic, Mechanical	NAND Flash, Electronic
Speed	Slower (Mechanical delay)	Much Faster
Durability	Fragile to drops	Highly Shock Resistant
Cost	Low	High
Application	Bulk storage, Backups	OS drive, Gaming, High-Perf

Topic: Memory-Mapped I/O

TOPIC OBJECTIVE:
To understand how peripheral devices communicate using the system memory address space.

TOPIC OUTCOME:
Students will be able to define Memory-Mapped I/O, explain the communication process, and differentiate it from Isolated I/O.

Introduction to Memory-Mapped I/O

Definition: A technique where I/O devices are assigned addresses within the same unified address space used for Main Memory.

- The CPU uses the **same instructions** (e.g., LOAD, STORE) to access RAM and I/O devices.
- CPU doesn't know if address 0xFFFF is a RAM chip or a Printer controller. It just writes to the address.
- **Advantage:** Simpler programming, no special I/O instructions needed.

Working of Memory-Mapped I/O

1. CPU executes a standard STORE instruction to an address.

2. Address Decoder looks at the address bus.

3. If Address < 0x8000 ➔ Routes data to RAM.

4. If Address > 0x8000 ➔ Routes data to Peripheral (e.g., Display).

Analogy: A city where houses (RAM) and businesses (I/O) share the exact same street numbering system.

Topic: Isolated I/O

TOPIC OBJECTIVE:
To understand Isolated I/O and compare it with Memory-Mapped I/O.

TOPIC OUTCOME:
Students will be able to define Isolated I/O, explain separate address spaces, and identify advantages.

Introduction to Isolated I/O

Definition: A technique where I/O devices use a separate, dedicated address space independent of main memory.

- Requires **special CPU instructions** (e.g., IN, OUT).
- Address 0x0010 in Memory space is completely different from Address 0x0010 in I/O space.
- **Advantage:** The entire memory address range is preserved purely for RAM.

Memory-Mapped vs Isolated I/O

Feature	Memory-Mapped I/O	Isolated I/O
Address Space	Single unified space	Two separate spaces
Instructions	Standard Memory (LOAD/STORE)	Dedicated I/O (IN/OUT)
RAM Capacity	Reduces available RAM space	Preserves full RAM space
Hardware Logic	Simpler decoding	Requires extra control lines

Memory Organization: Unit Assessment MCQ

Section A: Core Memory

Q1. Which memory is the fastest in a computer system?

A) DRAM B) SRAM C) Cache D) Register

Q2. Cache memory works on the principle of:

A) Locality of Reference B) Address mapping

Q3. DRAM is slower than SRAM because DRAM requires:

Section B: Mapping & I/O

Q4. In associative cache mapping, main memory blocks can be mapped to:

A) Any cache line B) Specific line only

Q5. Isolated I/O scheme uses which instructions?

A) LOAD/STORE B) IN/OUT

Q6. Memory-mapped I/O shares address space with:

A) System RAM B) Dedicated I/O ports

Section C: Advanced Concepts

Q7. Which mapping is conflict-free but expensive?

A) Direct B) Fully Associative

Q8. Write-through scheme in cache updates:

A) Only cache B) Both cache & main memory

Q9. Hit ratio is defined mathematically as:

A) Hits / Total Accesses B) Misses / Hits

Section D: Visual Analysis & High-Order Question 10

Q10. Analyze the performance trade-offs in Memory Hierarchy design:

As we go down the memory hierarchy pyramid (from Registers to Magnetic Disk/SSD):

A) Speed increases, capacity increases, cost per bit decreases
B) Speed decreases, capacity increases, cost per bit decreases
C) Speed decreases, capacity decreases, cost per bit increases
D) Speed increases, capacity decreases, cost per bit increases

Key Takeaway: Balancing access latency versus storage density determines system throughput.

Complete Unit Assessment

Section A (Short)

- Define Cache Hit.
- Define Locality of Reference.
- Differentiate SRAM/DRAM.

Section B (Long)

- Explain Cache Mapping techniques.
- Compare Memory Mapped vs Isolated I/O.
- Explain Memory Hierarchy.

University Exam Preparation

Important Diagrams to Practice:

- Memory Hierarchy Pyramid
- Cache Block Mapping (Direct/Set)
- HDD Internal Structure

Expected Viva Questions:

- Why is SRAM used in Cache?
- Why is SSD faster than HDD?
- What is the use of the Address Bus?

THANK YOU

MEMORY ORGANIZATION

Questions & Discussion

Presented By: Deepanshu Saini | Newton School of AI & Computing